INSTITUTE OF ENGINEERING & MANAGEMENT, KOLKATA
SALTLAKE
(School of University of Engineering and Management, Kolkata)

End Semester Examination, December - 2024

Degree: **B.C.A**, Semester: **Third**

Stream: **BCA**

Paper Code: **BCAAE301**

Paper Name: **Python Programming**

Full Marks:100 Duration:3 Hours

Part - A

Attempt 10 questions out of 15 questions

Each question carries 2 Marks (2 X 10)

1. What is an interpreter? Provide an example. **2**

2. Find the errors (if any) in the following segments **2**

   a) name = "Ram"
   name [2] = "b"
   print (name)

   b) i = input ('Get name')
   j = I + 10
   print (j)

3. What is the significance of a unique literal "None"? **2**

4. What is a namespace? **2**

5. What do you mean by the term "dead code" or "unreachable code''? **2**

6. What is the "pip" command in Python used for? **2**

7. Can mutable objects change their value but keep their id ()? **2**

8. How do loops work, and what are entry-controlled and exit-controlled loops? 2

9. What is function? List out the advantages of functions. 2

10. Define the "in" operator. 2

11. What is the output of the following operation? 2

```
set3 = {10,20,30,40,70,80}
set4 = {47,56,''24'',66,70,80}
Print (''set3 intersection set4 = \n'', set3 & set4)
```

12. What is the output of the following union operation? 2

```
set1 = {10,20,30,40}
set2 = {47,56,''24'',66}
set3 = set1.union(set2)
Print(set3)
```

13. What does the expression **info.get ("hobbies", None)** evaluates? 2

14. Write a program which receives either basic_pay and total_allowance or net_pay and print the net_pay using separate method and calculation of net_pay from basic_pay and allowance should be done using classmethod. 2

15. Explain different types of imports in Python with examples. 2

Part - B
Attempt 6 questions out of 9 questions
Each question carries 5 Marks (5 X 6)

16. What features of an interpreter does a compiler also have? What makes them unique from one another? 5

17. Take the customer's first name and last name and print them as last name first followed by the first name. 5

18. What does the term "automatic memory management" mean? 5

19. Create a Python script using for loop that will flip any user-entered word. 5

20. Write a program to use a for loop and a continue statement to go through all of the entered numbers one by one. The subsequent elements in the sequence are activated when a particular element in the sequence has a value of 16. 5

Given X = [23,43,56,78,16,9,7,12,14,15]

21. Write an iterative version function of the factorial of a number. 5

22. Write a Python program, which accepts an input number from the user and tells the user whether the number is even or odd. 5

23. What is the difference between tuples and lists in Python? 5

24. Explain the accessibility of variables in class based on the scope? 5

Part - C
Attempt 5 questions out of 8 questions
Each question carries 10 Marks (10 X 5)

25. What is operator precedence? List out the rules for operator precedence for arithmetic expressions. 10

26. An "attribute'' is a variable defined "inside" a module. On the other hand, a "method" is a function defined "inside" a module. How are these "attributes" and "methods" different from normal variables and functions, which are not imported from a module? 10

27. Create a code that uses an iterative for-else loop with continue to cycle through the fruit strings. Given sequence strings are juice = ["Pomegranate Juice", "Orange Juice", "Pineapple Juice", and "Grapes Juice"]. Healthiest Juice as the optional declared variable to be used in for loop. 10

28. Write a script which takes two numbers and finds their god (greatest common divisor) using Euclidean algorithm. 10

29. Write a program to iterate both lists simultaneously and display items from list 1 in original order and items from list 2 in reverse order. 10

Given

List 1 = [10, 20, 30, 40]

List 2 = [100, 200, 300, 400]

Expected output:

10 400

20 300

30 200

40 100

30. Given a pair of sets, generate a new set by collecting all of the elements from one Set that are absent from the other Set. 10

integer_set = {21,12,22,33,65,698}

magic_number_set = {2, 8, 20, 28, 50, 82, 126}

31. Write an OOP program which reads 2 points and performs following functions with users choice 10

a. Distance between 2 points.
b. Slope between 2 points
c. Checking whether the two points are inside/outside/on the circle $x^2 + y^2 = 16$

32. Write a program to implement single inheritance. 10

a) Create the parent class **Circle.** Initialise the constructor with the radius of the circle.
b) Define the method **get radius()** and **calc area()** to know the radius and area of the **circle.**

c) Create the child class named **Cylinder.** Initialise the value of the height within the constructor and call the constructor of the parent class to initialise the radius of the cylinder.
d) Finally, define the method **Calc area()** in the class **Cylinder** to calculate the area of the cylinder.

Note: Area of Cylinder = 2* pl * radius * height

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